

5.1 The natural and human causes of climate change

UNDERSTAND THESE TERMS

- Natural greenhouse effect
- Climate change
- Deforestation
- Greenhouse gases
- Enhanced greenhouse effect
- Fossil fuel

- 1 Draw a diagram to show the differences between the natural greenhouse effect and the enhanced greenhouse effect.
- 2 Sort the following into natural and human causes of climate change:
 - changes to Earth's orbit
 - volcanic eruptions
 - burning fossil fuels
 - deforestation
 - sunspot activity
 - agriculture
- 3 What is the difference between climate change and global warming?

4 What are three pieces of evidence used to measure climate change?

5 Fill in the gaps with the correct words below.

sea ice

paintings

ice cores

writings

- a** Scientists use _____ to study past temperatures by analysing gases trapped in ice.
- b** The position of _____ over time shows long-term climate changes.
- c** Historical records such as _____ and _____ provide evidence of past climate conditions.

When you are asked to predict and the question provides a source, such as a photo or a graph, remember to base your answer on the information provided.

6 Study Figure 5.1, which shows a cattle ranch in the Bolivian rainforest.



Figure 5.1: A cattle ranch in the Bolivian rainforest

Predict the likely factors contributing to climate change in Figure 5.1. [2]

35

- 7 Study Figure 5.2, which shows the total emissions of carbon dioxide and global temperature change between 1880 and 2018.

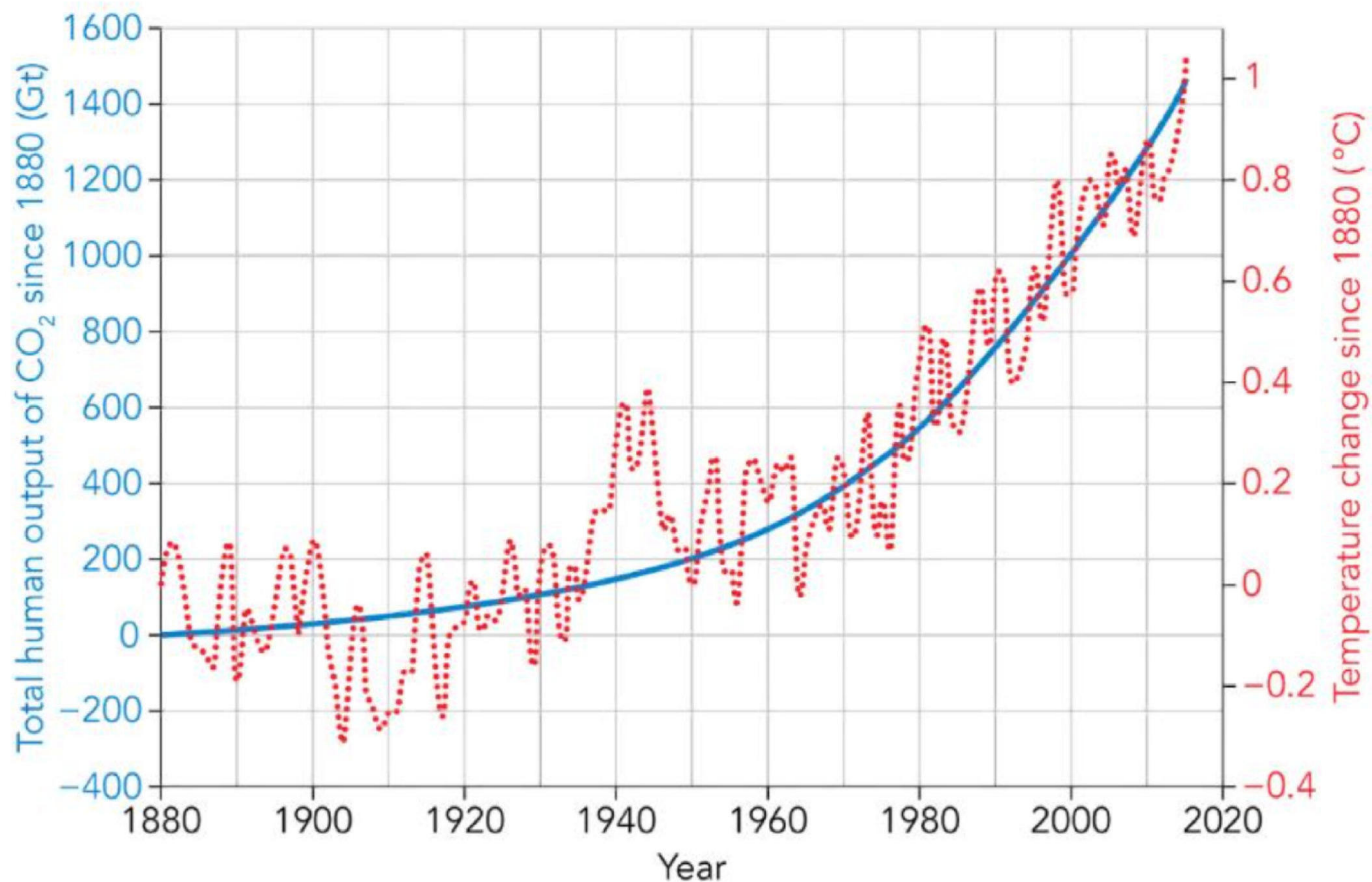


Figure 5.2: Graph showing carbon dioxide emissions and global temperature over time

Based on Figure 5.2, predict what will happen to global temperatures and carbon dioxide emissions in the future. [1]

When you are asked to assess, remember this means making a judgement by weighing up both strengths and weaknesses.

8 Assess the role of different greenhouse gases in global warming. [7]

9 Assess the extent to which humans are responsible for climate change. [7]

REFLECTION

What helps you to remember the difference between the enhanced greenhouse effect and climate change? For example, you could remember that warming in a greenhouse is linked to an increase in temperature only (enhanced greenhouse effect) and that change can be positive and negative (climate change).